GradientCIO

PORTFOLIO ATTRIBUTION COMPARISON

Northstar Attribution Comparison

Historical, forward and simulated views

PORTFOLIO
Northstar Total Fund

VIEWS
Three

AUDIENCE
Investment Committee

DATA AS OF
2026-10-08

OVERALL VIEW
Three compatible lanes compared

SCOPE REVIEWED
4 of 4 · Complete

01 Executive Comparison

Three compatible governed lanes have server-returned comparison metrics

OVERALL VIEW

Three compatible lanes compared

SCOPE REVIEWED · 4/4

BOTTOM LINE

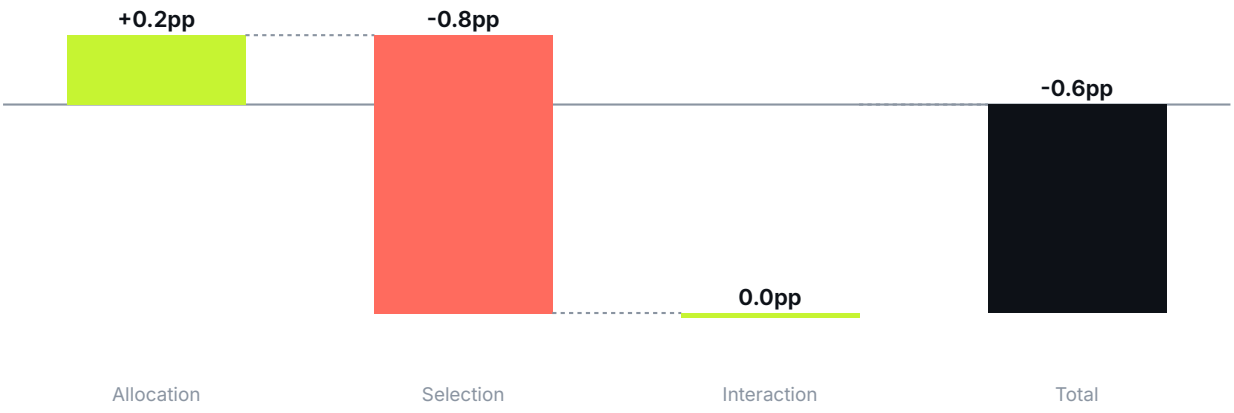
Historical, ex ante, and simulated effects are compatible, and governed comparison metrics are available S2 S3 S4 S5.

HISTORICAL LANE	EX ANTE LANE	SIMULATED LANE	COMPARISON COVERAGE
Available	**Available**	**Available**	**Complete**
Realized 12 months · 2025-07-01 to 2026-06-30 S2	1-year expected S3	1-year simulated · governed scenario S4	Server-returned comparison · compatible receipts S5

Three governed returned lanes and comparison metrics are available S2 S3 S4 S5.

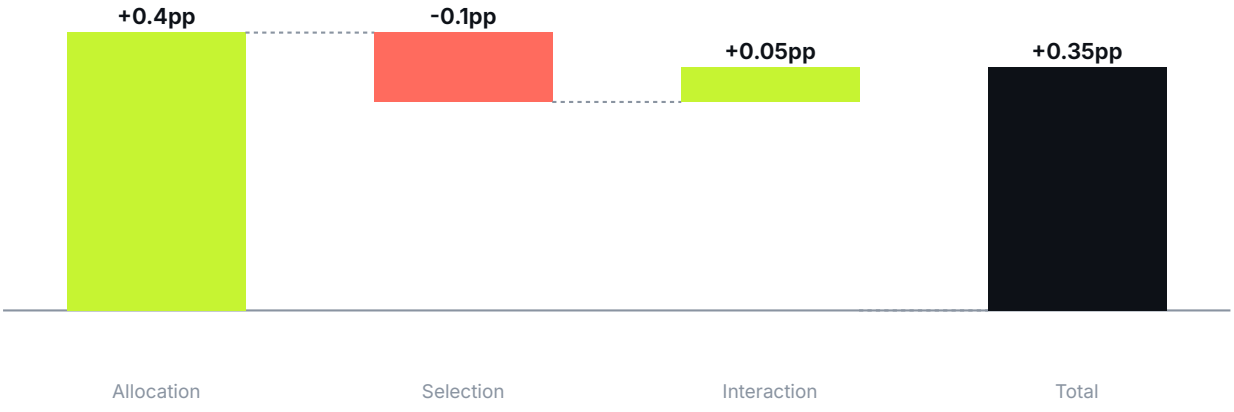
Returned evidence only; no local subtraction or accumulation.

REALIZED SUMMARY EFFECTS



Each plotted value is a returned summary field S2.

EXPECTED SUMMARY EFFECTS



Each plotted value is a returned summary field S3.

Key judgment — Historical

Observation: Historical effects are returned S2. Why it matters: Realized sources can be described. Uncertainty: One period may not persist. What would change the view: A later governed period.

Key judgment — Ex ante

Observation: Ex ante effects are returned S3. Why it matters: Expected sources can be described. Uncertainty: Inputs are assumptions. What would change the view: Refreshed governed inputs.

Key judgment — Comparison

Observation: Server-returned comparison metrics cover three compatible lanes S5. Why it matters: Cross-lane evidence is governed. Uncertainty: One comparison receipt limits time-series interpretation. What would change the view: A later receipt returning different lane signs.

02 Scope and Compatibility

Portfolio, benchmark, cohort and currency identifiers match

Compatible for governed three-lane comparison

The returned compatibility receipt matches portfolio, benchmark role, parent cohort and calculation currency S5.

Compatibility gate	● **available**	Four identifiers match S2 S3

03 Historical Attribution

Returned realized effects are shown without recalculation

Historical effects by segment

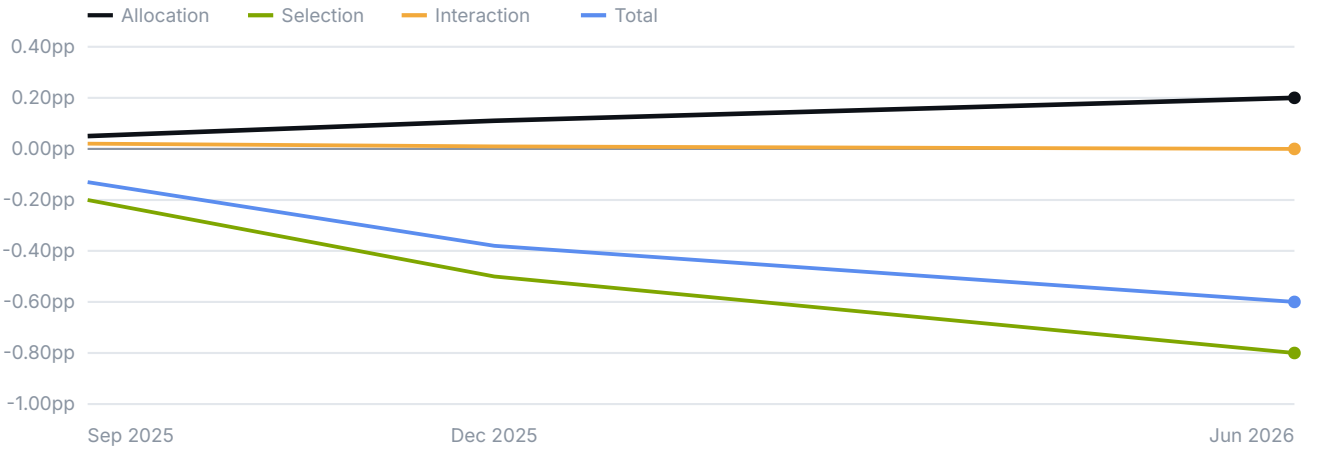
Not available — stacked values contain no positive total

HISTORICAL PORTFOLIO AND BENCHMARK AVERAGE WEIGHTS



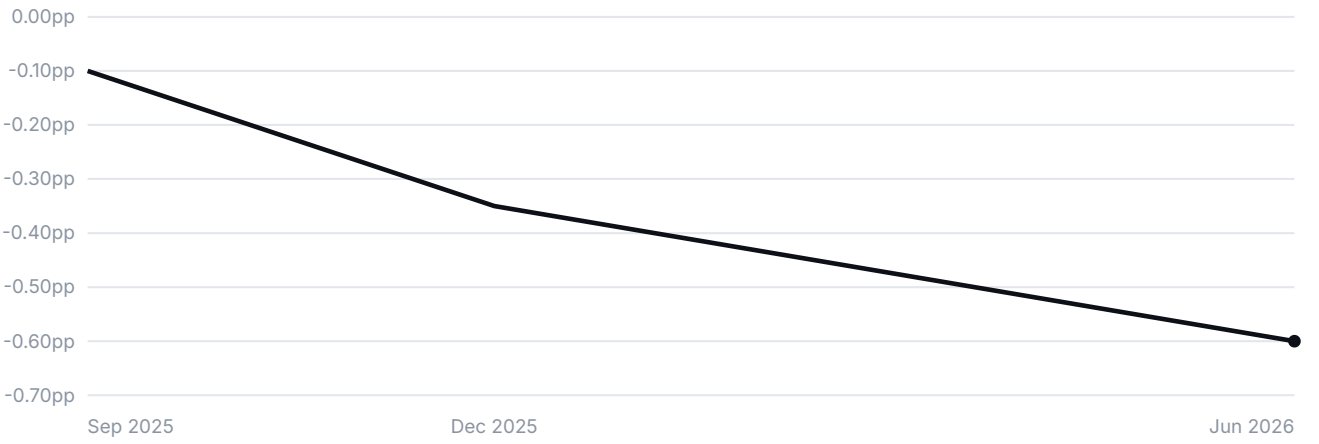
Returned arithmetic average beginning weights; no local difference S2.

CUMULATIVE LINKED EFFECTS



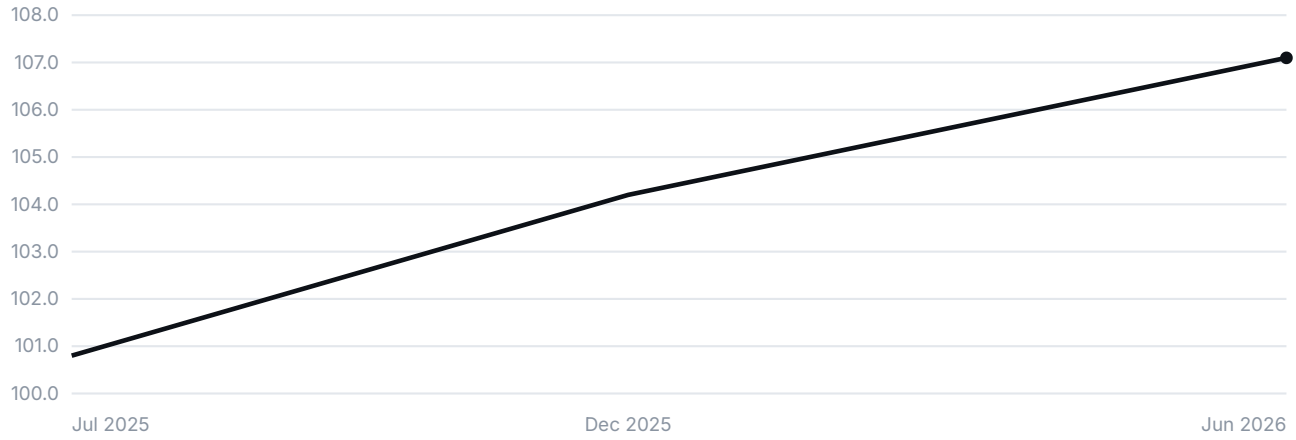
Server-returned cumulative linked fields; no local accumulation S2.

CUMULATIVE ALTERNATIVES SELECTION



Returned rows filtered to allocation_name=Alternatives; no local accumulation S2.

RETURNED CUMULATIVE GROWTH CONTEXT



Optional context rendered only from returned cumulative_growth rows S6.

Analysis — Selection leads detractors

Observation: Returned selection is negative S2. Why it matters: Selection leads realized detractors. Uncertainty: The evidence covers one linked period. What would change the view: A later historical result.

Benchmark and excess boundary

Returned diagnostics.entity_return_reconciliation keeps attribution-summary excess against the linked segment benchmark composite separate from historical-return excess against the policy benchmark entity S2.

04 Governed Ex Ante Attribution

Returned expected effects are shown as assumptions

EXPECTED EFFECTS BY SEGMENT

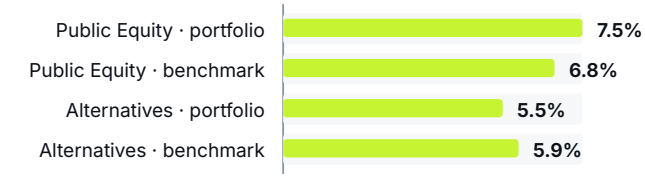


Returned expected allocation, selection and interaction by segment S3.

EX ANTE ASSUMPTIONS

assumption_set_name	**2026 Strategic CMA** S3
regime_name	**user-defined regime** S3
horizon	**1 year** S3
portfolio_return_source	**allocation_expected_return_base** S3
benchmark_return_source	**benchmark_expected_return_base** S3

EXPECTED PORTFOLIO AND BENCHMARK RETURNS



Returned expected returns grouped without local subtraction S3.

Analysis — Allocation leads expectations

Observation: Returned allocation is positive S3. Why it matters: Allocation leads expected effects. Uncertainty: Inputs are assumptions. What would change the view: A refreshed ex ante result.

05 Simulated Attribution

Governed simulation returns scenario effects

SIMULATED EFFECT BANDS



Server-returned scenario bands and selected values S4.

SCENARIO SELECTOR

SCENARIO ID	SCENARIO	SELECTED	BASIS
scenario-base-test	Base scenario	Yes	1-year simulated
scenario-down-test	Downside scenario	No	1-year simulated
scenario-up-test	Upside scenario	No	1-year simulated

Returned scenario selector; no locally invented scenarios S4.

Capability gate ● **available** Exact simulation capability was available before invocation S1 S4

Analysis — Scenario range returned

Observation: The governed simulation returned effect bands and a selected base scenario S4. Why it matters: Scenario context is explicit. Uncertainty: Results depend on governed scenario inputs. What would change the view: A revised simulation receipt.

06 Cross-Lane Comparison

Governed comparison returns compatible cross-lane metrics

RETURNED CROSS-LANE EFFECT MATRIX

	REALIZED	EXPECTED	SIMULATED
ALLOCATION	0.20	0.40	0.30
SELECTION	-0.80	-0.10	-0.20
INTERACTION	0.00	0.05	0.04

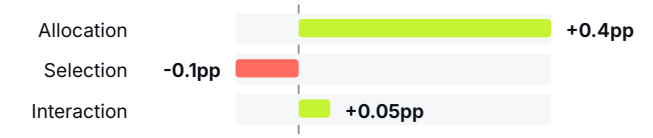
Server-returned three-lane effect matrix S5.

REALIZED RETURNED EFFECTS



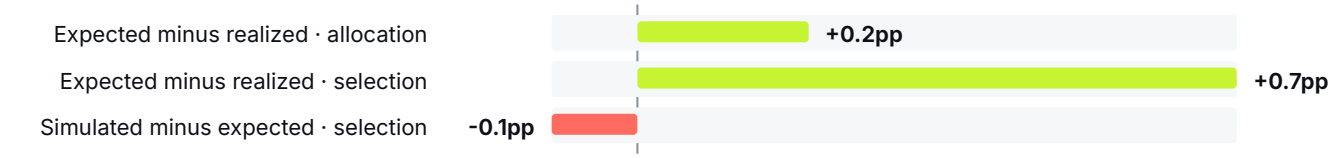
Displayed unchanged S2.

EXPECTED RETURNED EFFECTS



Displayed unchanged S3.

RETURNED LANE DIFFERENCES



Differences are returned by the comparison tool, not calculated here S5.

RETURNED CONCENTRATION



Server-returned concentration scores S5.

RETURNED PERSISTENCE

EFFECT	RETURNED LANES	PERSISTENCE STATUS
Allocation	Historical · Expected · Simulated	Persistent
Selection	Historical · Expected · Simulated	Persistent negative
Interaction	Expected · Simulated	Partial

Server-returned persistence labels S5.

Analysis — Governed comparison fields returned

Observation: The server returned differences, concentration and persistence for three compatible lanes S5. Why it matters: Cross-lane claims use governed fields. Uncertainty: One receipt limits time-series interpretation. What would change the view: A later comparison receipt returning different statuses.

07 Coverage and Limitations

Two live lanes and two capability gaps define coverage

Historical	available	Governed result S2
Ex ante	available	Governed result S3
Planned tools	available	Capability-gated results S4 S5

A Appendix — Sources and methods

Governed source inventory

TAG	EVIDENCE	BASIS	TOOL / VIEW	KEY PARAMETERS	AS OF	DATA SCOPE	VALIDATION	DIGEST
S1	Capability gate	gradient_tool	get_gradient_capabilities	illustrative sample	2026-06-30	Gradient-governed review evidence	passed	sha256:aaaaaaaaaaaa
S2	Historical attribution	gradient_tool	get_portfolio_attribution	illustrative sample	2026-06-30	Gradient-governed review evidence	passed	sha256:bbbbbbbbbbbb
S3	Ex ante attribution	gradient_tool	get_portfolio_ex_ante_attribution	illustrative sample	2026-06-30	Gradient-governed review evidence	passed	sha256:cccccccccccc
S4	Simulated attribution	gradient_tool	get_gradient_capabilities	illustrative sample	2026-06-30	Gradient-governed review evidence	passed	sha256:dddddddddddd
S5	Cross-lane comparison	gradient_tool	get_gradient_capabilities	illustrative sample	2026-06-30	Gradient-governed review evidence	passed	sha256:eeeeeeeeeeee
S6	Historical return context	gradient_tool	get_portfolio_historical_returns	fields=cumulative_growth; limit=100; illustrative sample	2026-06-30	Gradient-governed review evidence	passed	sha256:ffffffffffff

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

Full digests — S1: sha256:aaaaaaaaaaa